Name:______________________________ Period:______

Solving By Elimination Practice

1. $\begin{cases} r + s - t = 10 \\ r - 2s + t = 50 \\ 2r - 4s + 2t = 75 \end{cases}$

$r + s - \cancel{t} = 10$
$r - 2s + \cancel{t} = 50$
$2r - s = 60$

$(r + s - t = 10)2$
$2r - 4s + \cancel{2t} = 75$
$2r + 2s - \cancel{2t} = 20$
$4r - 2s = 95$

$(2r - s = 60) - 2$
$\cancel{4r} - \cancel{2s} = 95$
$-\cancel{4r} + \cancel{2s} = -120$
$0 \neq -25$

$\boxed{\text{No Solution}}$

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2. $\begin{cases} 2h + j + 5k = 57 \\ h + j + k = 16 \\ h + 6j + 2k = 35 \end{cases}$

$2h + j + 5k = 57$
$-1(h + j + k = 16)$
$-h - \cancel{j} - k = -16$
$2h + \cancel{j} + 5k = 57$
$h + \cancel{4k} = 41$
$-5h - \cancel{4k} = -61$
$-\frac{4h}{-4} = \frac{-20}{-4}$
$h = 5$

$-6(h + j + k = 16)$
$h + \cancel{6j} + 2k = 35$
$-6h - \cancel{6j} - 6k = -46$
$-5h - 4k = -61$

$5 + 4k = 41$
$-5 \quad -5$
$\frac{4k}{4} = \frac{36}{4}$
$k = 9$

$5 + j + 9 = 16$
$j + 14 = 16$
$-14 \quad -14$
$j = 2$

$\boxed{(5, 2, 9)}$

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3. $\begin{cases} 2x - 3y + z = 9 \\ x + 3y + 2z = -9 \\ 3y + z = -9 \end{cases}$

$2x - \cancel{3y} + z = 9$
$x + \cancel{3y} + 2z = -9$
$-2(3x + 3z = 0)$
$3(2x + 2z = 0)$

$-\cancel{6x} - \cancel{6z} = 0$
$\cancel{6x} + \cancel{6z} = 0$
$0 = 0$

$\boxed{\text{IMS}}$

$2x - \cancel{3y} + z = 9$
$\cancel{3y} + z = -9$
$2x + 2z = 0$